A prospective assay for hypothesis-space expansion in AI systems

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Abstract

Scientific discovery can require changing not only a hypothesis but the language in which hypotheses are expressed. We introduce a prospective, replayable assay for hypothesis-space expansion. Candidates must be structurally outside a frozen incumbent language, fit observations, commit to a discriminating intervention before outcome reveal, outperform the incumbent predictor and survive independent falsification. Typed high-level proposals succeeded in 142/400 synthetic worlds, while generic local rewrites succeeded in 400/400 known-family and 100/100 held-out worlds. Crucially, code-path ablation reproduced rewrite-search verdicts without model output, showing that scientific content was supplied by deterministic scaffolding rather than the language model. Legacy autonomous runs failed before execution. A separately frozen, grammar-valid interface instead yielded executable plans in 3,939/4,608 opportunities and validated escapes in 15/96 worlds, concentrated in two families. The assay therefore reveals both scaffold capability and interface-sensitive model attribution while establishing measurement validity, not general scientific-discovery competence beyond these structural families or model settings.

Introduction

Scientific change can alter the representational vocabulary in which hypotheses are stated[1-3]. Computational creativity and constructive induction distinguish exploration within a space from transformations of that space[4-9]. Symbolic regression, program search and language-model-guided systems discover equations, algorithms and hypotheses[10-17], while multi-agent systems automate broader workflows[18-20].

Recent work now explicitly targets expanding principle or model spaces. PiEvo treats discovery as optimization over an evolving principle space across four benchmarks[21]. Model Discovery Agent couples a language-model proposer to Bayesian experiment design in an open model setting spanning physics, chemistry and biology[22]. HypoArena evaluates prospective hypothesis discovery in 988 cases across six domains and 15 frontier models[23]. EvoSCM evolves structural causal models through intervention and prospective prediction[24]. These studies broaden the scientific tasks and models under evaluation. A complementary measurement problem remains: how can an evaluation prove that a candidate left a specified incumbent hypothesis language, charge it a prediction before seeing the answer, and separately identify which system component caused the escape?

Here we operationalize bounded hypothesis-space expansion. We freeze an incumbent representational language, generate executable candidates and require a six-gate verdict: incumbent adequacy (J0), canonical structural non-membership (J1), candidate adequacy (J2), a discriminating committed prediction (J3), prospective intervention gain (J4) and independent falsification survival (J5; Fig. 1). Candidate prose, model confidence, embedding distance and semantic novelty do not enter the verdict.

Our experiments validate the assay and reveal a component-attribution limit. Typed proposals and generic compositional search cross the frozen language reliably, whereas fixed-space alternatives do not. However, code-path ablation shows that C3's representations, fitted equations, ranking and interventions were deterministic; removing model output leaves all 500 jump and 300 control verdicts unchanged. The audit demonstrates how a system-level discovery benchmark can misattribute successful scientific content to a model when deterministic scaffolding supplies the decisive representation, fit and intervention. The contribution is therefore the prospective assay, the typed search result and a method for auditing causal system attribution-not evidence that a language model produced the successful escapes.

Table 1 | Position relative to adjacent discovery evaluations

Evaluation	Exec	Frozen	Canonical out-of-space test	Prospective	Falsify	Attribution / replay	Breadth
Hypothesis Search / HypoGen	partial	no	no	no	no	no	language tasks
POPPER	partial	no	no	yes	yes	limited	literature
FunSearch	yes	skeleton	no	feedback	tests	proposer/evaluator	mathematics
PiEvo	yes	evolving	no	varies	varies	proposer/search	4 benchmarks
Model Discovery Agent	yes	open set	no	Bayesian	predictive	proposer/inference	3 sciences
HypoArena	judged	no	no	context	rubric	no	988 / 6 / 15
EvoSCM	yes	evolving	no	active	prospectiv e	evolution/selection	causal systems
This work	yes	yes	canonical	locked	exact	factorial / exact	9 families / 2-checkpoint sensitivity

Results

A prospective criterion for hypothesis-space expansion

Figure 1 | A prospective assay for bounded representation escape

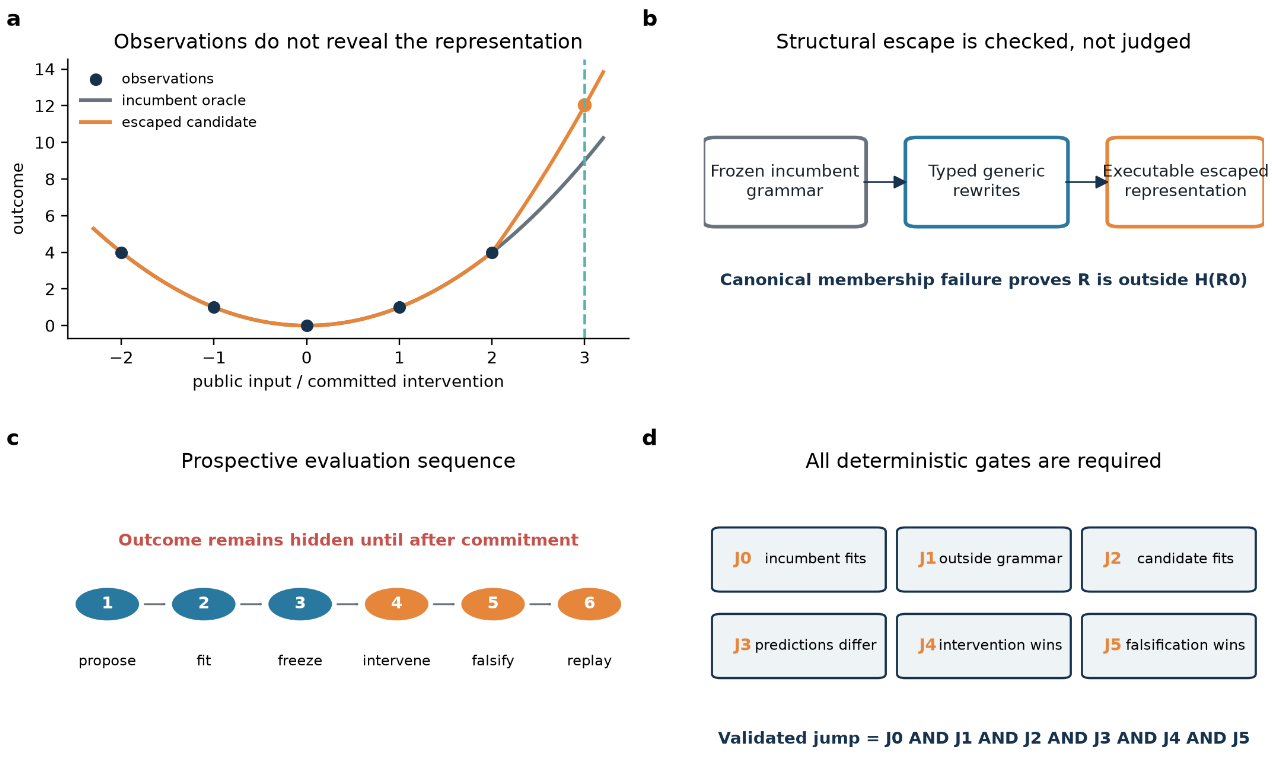

Figure 1 | The assay combines canonical structural non-membership with a prospective intervention and independent falsification. Panel a is schematic; panels b-d summarize the registered pipeline.

Each synthetic world contains observations compatible with an incumbent model, a frozen typed-graph language, hidden intervention outcomes and independent falsification cases. The best incumbent predictor is computed exactly or by bounded exhaustive search. Before any intervention outcome is revealed, a candidate representation, executable expression, selected action, prediction and split hash are committed. A validated jump is the conjunction $J0 \wedge J1 \wedge J2 \wedge J3 \wedge J4 \wedge J5$ (Fig. 1). This design turns "leaving a hypothesis space" into a replayable event rather than a judgement about wording or apparent novelty.

Typed proposals outperform fixed-space alternatives

Figure 2 | Typed proposals and their gate attrition

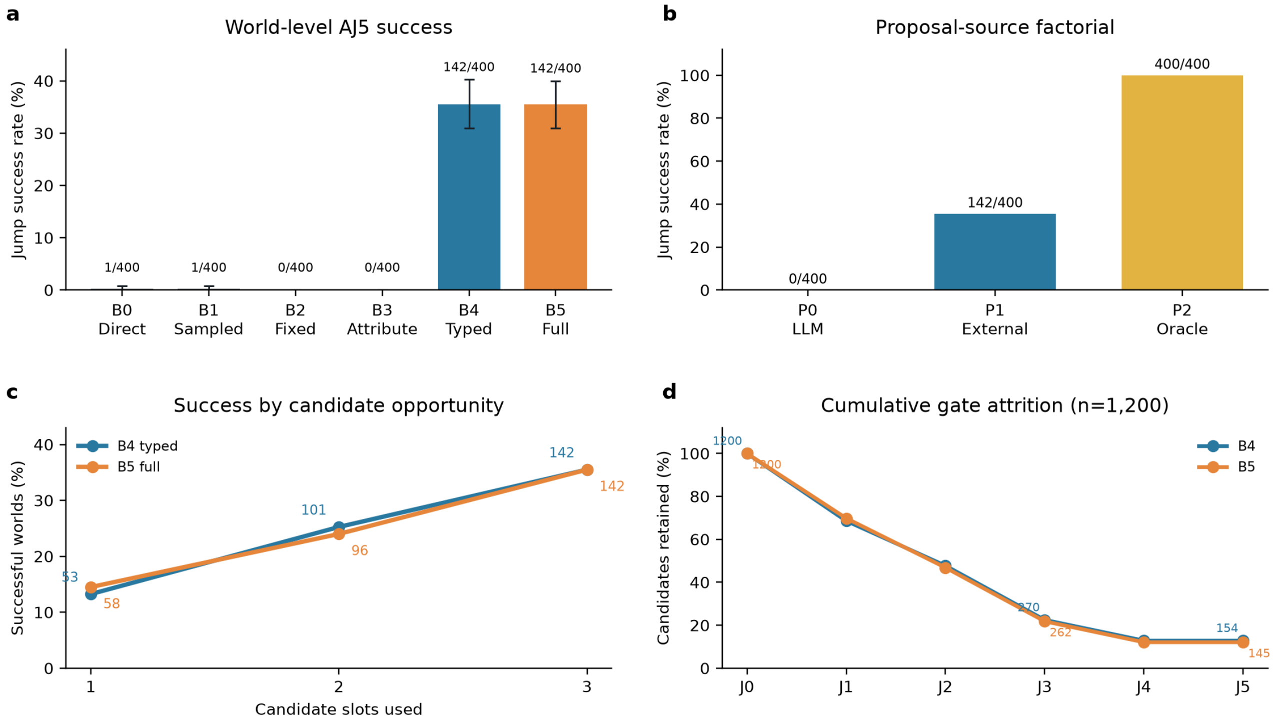

Figure 2 | AJ5 world-level jump success. Error bars are registered family-stratified bootstrap 95% intervals. Counts are successful worlds; n=400 per jump condition. All conditions recorded 0/200 false jumps.

In AJ5, direct (B0) and sampling-matched (B1) model proposals each succeeded in 1/400 worlds; fixed-space reasoning (B2), attribute-only mutation (B3) and value-only mutation succeeded in none. External typed proposals (B4 and B5) each succeeded in 142/400 (35.5%; Fig. 2a). All eight registered contrasts with B0-B3 were positive after Holm correction (adjusted P<=4.00x10-4).

Holding the downstream path fixed, model-proposed representations (P0) succeeded in 0/400 worlds, external typed proposals (P1) in 142/400 and oracle representations (P2) in 400/400 (Fig. 2b). The system can execute a supplied changed representation, but this does not establish model necessity after deterministic fitting because the second call received a fitted expression and maximum-separation intervention that were programmatically enforced.

Gate attrition clarifies the result (Fig. 2c). From 1,200 jump-world candidates each, B4 retained 823 at J1, 573 at J2, 270 at J3 and 154 at J4-J5; B5 retained 838, 562, 262 and 145. Controls still produced 112 and 110 candidates through J3, but none passed J4: prospective outcomes, not J0-J3 alone, eliminated these apparent escapes.

Generic rewrites compose into validated representations

Figure 3 | Generic rewrites compose into validated representations

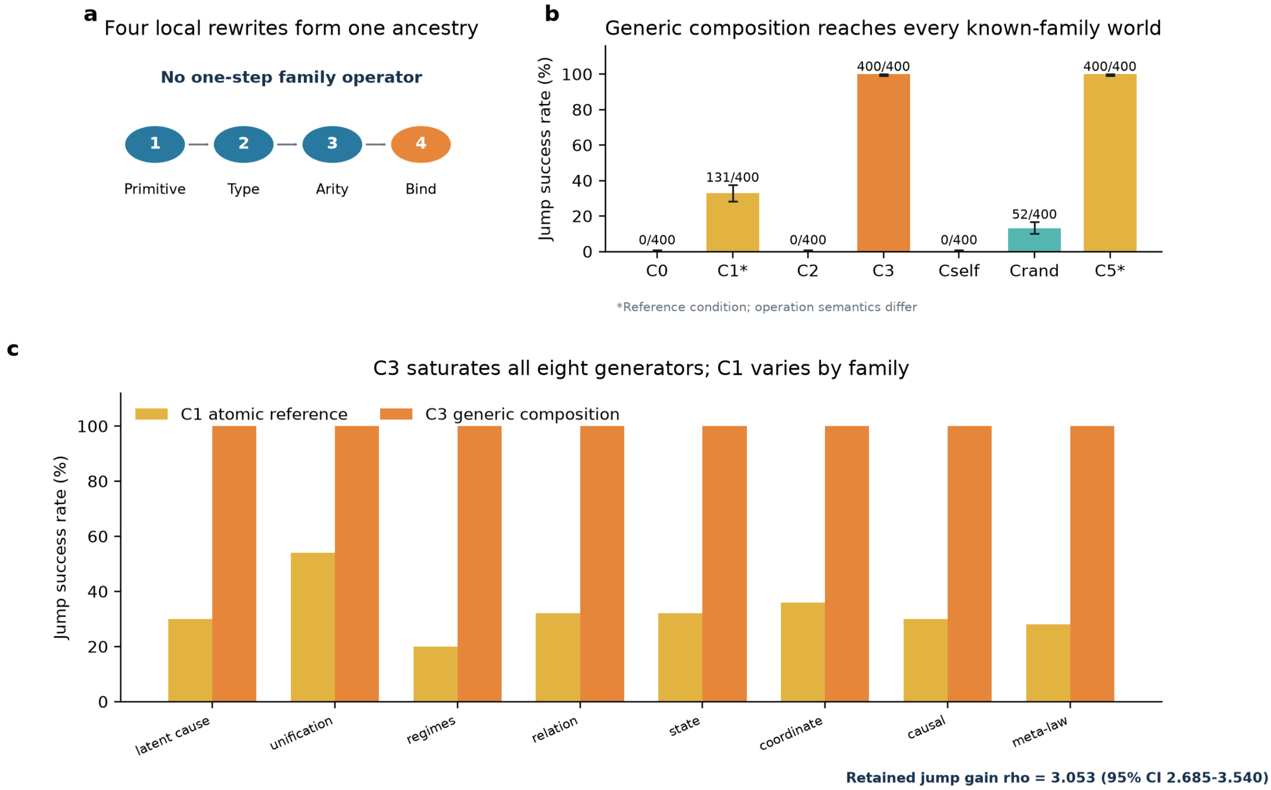

Figure 3 | CJ5 known-family reconstruction. Error bars in panel b are Wilson 95% intervals; n=400 worlds per condition. C1 and C5 are reference conditions with different operation semantics. Per-family panels contain 50 worlds each.

CJ5 replaced nine high-level mutations with 29 local graph and syntax-tree rewrites. Node addition, type, observability, arity and argument binding were separate steps; no primitive accepted a family label, target distance or outcome. C3 traversed 48 four-step branches, ranked candidates without outcomes and retained three.

Across eight known families (n=400 worlds), fixed-space search (C0) and depth-one alternatives (C2) succeeded in none. The atomic reference (C1) succeeded in 131 (32.75%), random four-step paths (C_rand) in 52 (13.0%) and C3 in all 400 (Fig. 3). C3 exceeded C_rand by 0.87 (95% CI 0.845-0.895; adjusted P=3.00x10-4). Its family-wise saturation demonstrates within-generator reliability and assay separability, not eight independent replications or a frontier benchmark.

Code-path ablation attributes C3 success to deterministic scaffolding

Figure 4 | Component attribution, transfer and model sensitivity

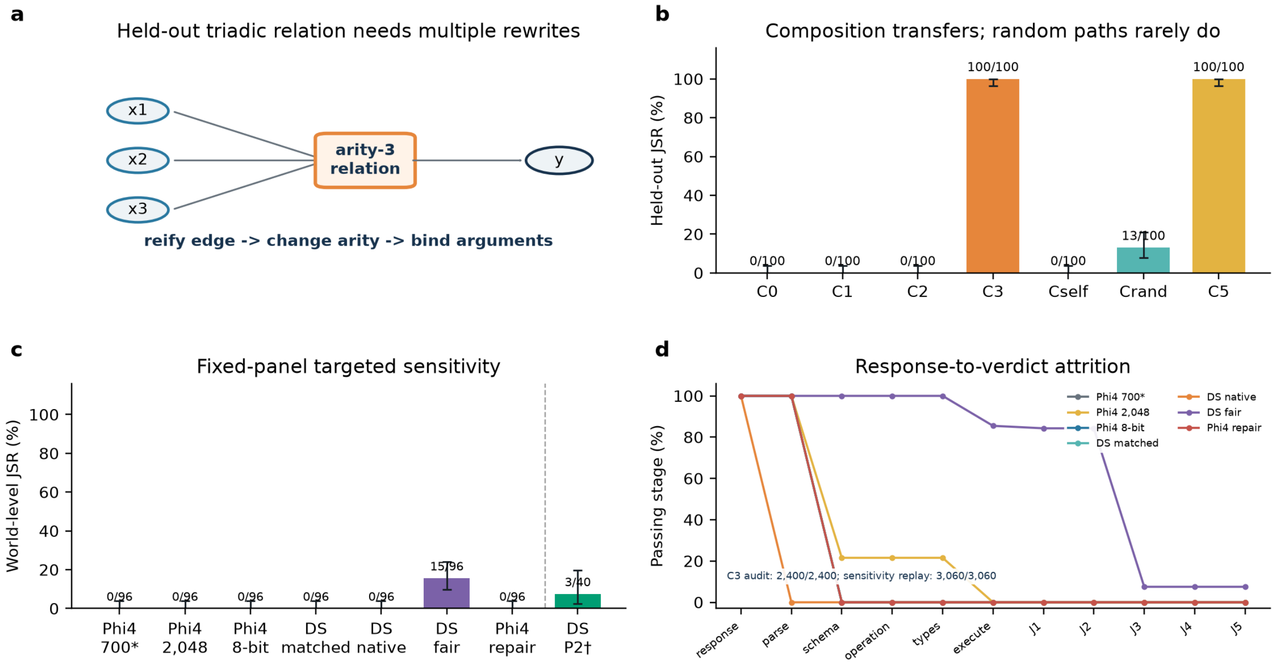

Figure 4 | Component attribution, held-out transfer and the minimal targeted sensitivity extension. Panel c reports the fixed 96-world paired panel, except the predeclared n=40 supplied-representation control (dagger); the asterisk marks the fixed historical slice of the original n=400 Phi-4 confirmatory population. The Phi-4 2,048-token condition changes only the historical completion cap and is not compute-matched. Error bars are Wilson 95% intervals. In panel d, response and structural-validity stages use their explicitly reported response or effective-plan denominators, whereas J1-J5 use executable self-proposed candidates; the connecting lines aid stage ordering but do not imply a common denominator across that boundary. Historical parse validity denotes legacy object extraction, not strict whole-response JSON. No candidate-level significance tests were performed.

Inspection of the frozen C3 path showed that representation, expression, ranking and intervention selection occurred deterministically before the model response. Response fields for representation, expression and intervention were overwritten; only an explanation remained, and explanations do not enter J0-J5. We therefore replaced both model calls with a valid empty explanation and recomputed every theory, commitment and verdict without inference.

All 2,400 candidate verdicts matched the archive. The replay retained 500/500 jump-world successes and 0/300 control false jumps (Fig. 4). Phi-4 was therefore not causally necessary for C3. The perfect result belongs to the typed deterministic search-and-evaluation scaffold. More generally, the audit shows why scientific-discovery evaluations need code-path attribution: system success alone does not identify which component supplied the scientific content.

Legacy sensitivities expose proposal-interface attrition

C_self scored 0/400, but offline reconstruction showed that this cannot isolate proposal quality. All 1,200 responses were non-empty and JSON-extractable, yet none contained a valid outer plans list; 0/19,200 registered opportunities reached execution or J1. Every response reached the 700-token cap, and none was strict whole-response JSON. The historical failure therefore conflates representation proposal with truncation and serialization.

Direct contract inspection revealed an additional confound. The prompt enumerated operators but omitted their exact argument keys: for example, the executor expects node, other and kind, whereas all 288 matched DeepSeek proposal calls used at least one of source, target or type. None was strict whole-response JSON. The permissive legacy extractor could also recover an inner object from truncated outer JSON, so "JSON extractable" did not imply a valid top-level plan. The registered interface was therefore unsuitable for isolating conceptual proposal quality.

We used a fixed, outcome-blind panel of 96 historical worlds (12 per family) to test precision, budget and model-interface alternatives without replacing the original n=400 confirmation (Fig. 4). Every legacy-interface autonomous condition remained at 0/96 (Wilson 95% interval 0-3.8%). All five paired JSR differences were 0.000: longer-budget and 8-bit Phi-4 versus the historical slice, matched DeepSeek versus that slice, native versus matched DeepSeek, and validator-only repair versus unrepaired 8-bit Phi-4. These are targeted sensitivities, not candidate-level tests or a new confirmatory population.

Attrition remained predominantly pre-executable. On the panel, longer-budget Phi-4 produced 993/4,608 schema-valid opportunities but none executable; Phi-4 8-bit and its one-repair condition produced no schema-valid plans. Matched DeepSeek also produced none: 4,480 opportunities had a non-list plans field and 128 lacked plans. In the full longer-budget populations, only 9/19,200 known-family opportunities and 0/4,800 held-out opportunities were executable, with no successful world. Precision, budget, repair and checkpoint substitution therefore did not recover an autonomous jump under this interface.

Native DeepSeek also scored 0/96, but for a different interface reason: all 576 calls returned reasoning text, no separate answer reached the parser and 518 hit the 4,096-token cap. This does not isolate conceptual proposal quality from reasoning-to-answer serialization. In the supplied-representation control, 4/120 model-authored candidates were parse-valid and executable and 3 passed J1-J5, yielding 3/40 successful worlds (7.5%; Wilson 95% interval 2.6-19.9%); 116 calls hit the cap. The checkpoint can therefore use a supplied correct representation occasionally, but downstream reliability remains poor. These runs do not establish model-independent conceptual incapacity.

Grammar-valid serialization reveals family-concentrated escapes

We next froze a single fair-interface DeepSeek sensitivity on the same 96 worlds. Separate deliberation and answer budgets plus a strict plan schema guaranteed grammar and parser-level argument syntax but supplied no truth, fit, intervention outcome, gate value or semantic repair. All 4,608 plan opportunities were strict whole-response JSON and schema-valid; 3,939 (85.5%) were dynamically executable. At least one plan was executable in 280/288 candidate slots. Among those selected candidates, 236/280 passed J1-J2, 21/280 passed J3 and the same 21 passed J4-J5. These yielded 21 validated candidates and 15/96 successful worlds (15.6%; Wilson 95% interval 9.7-24.2%). Relative to each of the historical Phi-4 slice, matched DeepSeek and native DeepSeek, 15 worlds changed from failure to success and none changed in the opposite direction.

Success was concentrated in two families: meta-law succeeded in 9/12 worlds and unification in 6/12; the other six families each scored 0/12. All 288 deliberation calls reached the 4,096-token cap, whereas all 288 serialization calls stopped before their separate cap. Grammar-valid serialization therefore exposed autonomous structural escapes that the legacy interfaces could not measure, while the family concentration and sharp attrition from J2 to J3 show that executable syntax did not make the task generically easy. This result supports interface-sensitive capability under the tested checkpoint, operator language, budget and world panel-not broad autonomous scientific-discovery competence.

Transfer to one held-out structural family

The second study reserved triadic_relation_reification from development, pilot inference and known-family confirmation. C3 succeeded in 100/100 held-out worlds, C_rand in 13/100, and C0, C1, C2 and C_self in none (difference from C_rand 0.87, 95% CI 0.80-0.93; adjusted P=3.00x10-4).

The holdout is limited: an earlier family used binary property-to-relation change, and the generic language already included reification. Only the arity-three structure, product mechanism, generator and instances were new. One adjacent family cannot support broad external generalization.

A worked prospective escape

Figure 5 | One complete prospective escape

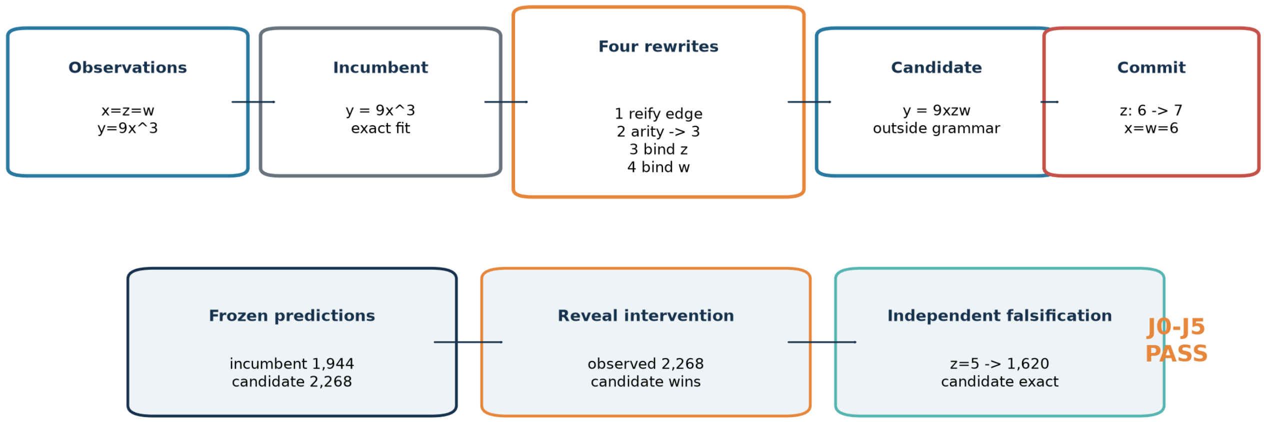

Figure 5 | Worked held-out example. Correlated observations make the cubic incumbent and triadic candidate observationally identical. A committed intervention separates them before outcome reveal, and an independent case falsifies the incumbent.

In one held-out world, observations with x=z=w gave $y=9x^3$, so the incumbent $y=9x^3$ and candidate y=9xzw both fit exactly. Four generic edits reified the incumbent prediction edge, changed the new relation to arity three, and bound z and w (Fig. 5). Before outcome reveal, the assay committed to setting z from 6 to 7 at x=w=6. The incumbent predicted 1,944; the candidate predicted 2,268; the revealed outcome was 2,268. Independent falsification at z=5 yielded 1,620, again exact for the candidate. The representation failed the frozen language on relation count, type, arity bindings and result structure, and passed J0-J5.

Specificity and replay

The zero false-jump count requires qualification. In C3 controls, 800 of 900 candidates passed J1, 567 passed J2 and 283 passed J3, but none passed J4. The null was therefore non-trivial through the prospective commitment gate. Nevertheless, because no-jump truth is generated inside the incumbent language and the incumbent oracle is exact, an outside candidate cannot strictly beat it under noiseless evaluation. The observed 0/300 is best interpreted as a structural-specificity check under the exact simulator, not an empirical false-positive estimate for noisy science; the Wilson upper bound is descriptive only.

All 10,800 AJ5 and 16,800 CJ5 selected candidates replayed exactly, including 35,533 CJ5 ancestry records. No confirmatory world was excluded and no shard was rerun.

Discussion

This work contributes an assay for a specific event: an executable candidate leaves a frozen hypothesis language and pays for that move with a prospective prediction that beats the best incumbent and survives independent falsification. Canonical non-membership, outcome-before-commitment ordering and exact replay distinguish this event from judged novelty or retrospective fit (Table 1).

The code-path audit is itself a methodological result. It demonstrates that a system-level discovery benchmark can assign successful scientific content to a language model even when deterministic scaffolding supplies the decisive representation, fit, ranking and intervention. Conversely, the legacy autonomous zero rates were dominated by pre-executable attrition despite precision, budget, checkpoint and repair sensitivities. A grammar-valid interface restored executability and revealed validated model-proposed escapes, but only in two of eight families. End-to-end failure can therefore underestimate capability when content never reaches the evaluator, just as end-to-end success can overattribute capability when decisive content comes from a scaffold. Claims of model-driven representation change require both provenance and survival of model-generated scientific content into the committed theory.

External validity is the principal limitation. The worlds are synthetic and noiseless; C3 is saturated; search strata and generators were co-designed; and the operator language embeds graph, function, relation and binding priors. The original confirmation used one model; the extension sampled only one additional checkpoint on a fixed subset, and its native condition was not compute-matched. Only one conceptually adjacent family was held out. World seeds quantify within-generator reliability. The higher-level generalization units are nine structural families, only one of which was held out, so small world-level P values do not substitute for structural breadth.

A decisive extension should freeze the current assay and add independently authored families, distractor primitives, deeper compositions, stochastic observations, partial observability and at least one recognizable mechanistic simulation. Broader open and proprietary model comparisons could then use the tested separation of reasoning and grammar-valid answer stages, while reporting dynamic executability and J1-J5 rather than treating format compliance as scientific success. These experiments are proposed, not part of the present evidence.

The conclusion is narrow but durable: hypothesis-space expansion can be measured prospectively and replayed exactly; typed deterministic search can produce it in controlled worlds; and causal component auditing can reveal when apparent capability or incapacity instead belongs to the scaffold or interface.

Methods

Study design and protocol freezing

AJ5 and CJ5 protocols were frozen in Git before their reported confirmatory model calls. AJ5 uses commit 895ebb9118ffd0046825b88868621f2a70f69f61. CJ5 uses 65f20874e16bddf8a7ae36996395ff52b27153b7, with a documented correction at 7ecb977 and held-out unlock at 27ee542. The commits are publicly retrievable but unsigned, and no independent registry or transparency-log timestamp was located. We therefore describe the studies as prospectively specified and commit-frozen, not formally preregistered. Pilot seeds were excluded; no confirmatory result changed families, operators, thresholds, budgets or seeds.

Model and inference

Both studies used frozen microsoft/phi-4 revision 2db69c1c3e91a05d2c64a3185acfbaf36f744e25, vLLM 0.10.2, dynamic bitsandbytes 4-bit quantization, a 4,096-token context, temperature 0.2, top-p 0.95 and completion cap 700. AJ5 sampling proposals used temperature 0.7. Each request had a deterministic seed. No fine-tuning or cross-world adaptation occurred.

Minimal targeted sensitivity extension

The original Phi-4 artifacts were preserved at commit ae1ede683fdef09f2bf60f6e1052b60394ad6cf8, tag nmi-phi4-frozen-2026-09 and archive branch nmi-phi4-frozen-archive-2026-09. A separate extension protocol, panel, model configurations and generation code were frozen at commit 320eb29b33ddaed596b0fe7b3f1d5895c706f311 and amended only for operational execution at f846c89287e379fe313551c47765c24f2abf4959. New outputs used the nmi_minimal_sensitivity_v1 namespace and could not overwrite historical files.

An outcome-blind SHA-256 ranking selected 12 of the 50 historical CJ5 seeds before any new model call: 30014, 30012, 30029, 30025, 30011, 30023, 30000, 30032, 30001, 30037, 30002 and 30015. Applying the same seeds to all eight known families produced a fixed 96-world paired panel. The positive control used the first five selected seeds in each family (n=40). This extension is a targeted sensitivity analysis, not a replacement confirmatory population.

The frozen run matrix contained Phi-4 8-bit C_self on 96 worlds; DeepSeek matched C_self with reasoning_effort=none on the same worlds; DeepSeek native C_self with reasoning_effort=max; and a supplied-correct-representation DeepSeek positive control. The DeepSeek service identified the checkpoint as deepseek-ai/DeepSeek-V4-Flash-Vision-Exp, revision 86f746b36186f0e567729a5c06a8c918caba82a9, served as deepseek-v4-flash-vision-exp by vLLM 0.25.2.dev0+g752a3a504.d20260714. We do not equate this checkpoint with any other DeepSeek release. The matched condition retained the historical 700-token output cap; the native and positive-control conditions froze a 4,096-token output cap. Reasoning text was returned separately in message.reasoning; a separate reasoning-token count was recorded only if exposed by the service.

A separately prospectively frozen PHI-BUDGET-SENSITIVITY condition retained the historical Phi-4 revision, 4-bit runtime, prompt semantics, primitive vocabulary, worlds, three candidate slots, 16 plans per slot, J0-J5 gates, interventions and hidden-information boundaries. Its only scientific change was increasing the predeclared completion cap from 700 to 2,048 tokens; it did not add representation attempts, candidates or interventions and is not compute-matched to the historical condition. The source protocol was frozen at commit 4606413 and tag nmi-extension-v1-protocol-freeze. Complete known-family (n=400) and held-out (n=100) shards were incorporated under outcome-blind amendment NMI-MIN-SENS-V1-AMENDMENT-002 before analysis. The fixed 96-world panel provides the primary paired budget comparison, while the complete populations are reported descriptively.

Worlds, primitive vocabulary, three candidate slots, 16 four-step plans per slot, representation opportunities, J0-J5, prospective interventions and hidden-information boundaries were unchanged. The matched condition received no additional semantic information. The positive control fixed only the correct representation; the model still authored the executable expression, explanation and intervention selection, so success could not be created by overwriting those fields. Because the offline historical cascade had 100% pre-executable attrition, exceeding the frozen 25% trigger, one additional Phi-4 8-bit condition permitted exactly one replacement repair response per structurally invalid slot. Validator feedback named only syntax, schema, operation, reference, type or arity errors and exposed no truth, target distance, outcome or gate value.

Fair-interface sensitivity

After auditing the frozen responses, we specified one additional DeepSeek condition on the identical 96-world panel. Each of three slots retained 16 independent four-step plans and therefore 48 representation opportunities per world. The first call reserved 4,096 tokens for native reasoning; a second call received that model-produced deliberation and a complete parser-level operation syntax, used reasoning_effort=none, and reserved 4,096 tokens for a compact grammar-constrained plan object. These two calls replaced the original proposal-plus-explanation allocation, preserving six calls per world. The schema guaranteed only JSON structure, plan count, depth, operator names and argument keys/types. Dynamic references, executable ordering and scientific usefulness remained model responsibilities.

The serializer received no fitted expression, truth, target distance, intervention outcome, gate value or semantic repair. Worlds, primitive vocabulary, candidate slots, opportunities, deterministic downstream fitter and intervention selector, hidden-information boundaries and J0-J5 were unchanged. The protocol and code were frozen at commit b6e1561 and tag nmi-fair-interface-v1-protocol-freeze. A subsequent operational amendment partitioned families into four disjoint shards without changing calls or scientific settings; an interrupted 31-call throughput pilot was archived and excluded before the formal shards began. Concurrent shard launch then exposed a four-sequence server limit: three timeout-only shard directories contained no returned response or result table. We archived and excluded them, froze a second operational amendment at commit 6bab5fb and tag nmi-fair-interface-v1-sequential-shards, and scheduled the unchanged shards sequentially.

Worlds, representations and search

Generators returned public observations, redacted actions, a frozen incumbent LanguageSpec, hidden truth and independent intervention and falsification cases. AJ5 used eight families and 50 jump seeds per family; CJ5 used 50 new seeds per family. No-jump controls used 25 seeds per family in each phase. CJ5 added 100 held-out jump and 100 control seeds. Representations were canonical typed graphs with executable fitted programs.

AJ5 tested direct model output, model sampling, fixed-space reasoning, attribute-only mutation and two typed high-level proposal conditions. CJ5 used 29 generic local rewrites. C3 traversed 48 four-operation branches and retained three candidates by a deterministic score. C_rand sampled 48 four-step paths. C_self requested 16 plans in each of three slots; invalid plans were not repaired.

Prospective evaluation

J0 required incumbent observation MSE<=10-12; J1 required DSL validity and frozen-language non-membership; J2 required candidate observation MSE<=10-12; J3 required candidate-oracle prediction separation>=0.5 on the committed action; J4 required intervention MSE improvement>0.1; J5 required falsification MSE<=10-12 and improvement>0.1. Candidate, prediction, action and split hashes were committed before simulator evaluation.

Statistics and component audit

The world was the replicate. AJ5 used 10,000 family-stratified paired bootstrap replicates; CJ5 used the same bootstrap for effects and paired sign-flip tests. Holm correction was applied within prospectively specified comparison families at $\alpha=0.05$. Candidate rows were used only for attrition. Family-level distributions are reported descriptively because eight known families plus one held-out family do not support a stable population-level family inference.

The post-hoc component audit reconstructed all C3 candidates from archived deterministic representations, fitted expressions and selected interventions, replacing the model-derived explanation with an empty string. It then regenerated each world, froze a new commitment and recomputed J0-J5. No model inference was run.

For the targeted sensitivity analysis, exact world counts, JSR and Wilson 95% intervals are primary. Comparisons use paired world transitions and paired JSR differences on the identical 96-world panel; family rates are descriptive. The full n=400 known-family and n=100 held-out Phi-4 budget populations are supplementary descriptive sensitivities, and the n=40 positive control is shown separately. No candidate-level significance tests are performed and P values are not used as primary evidence. Response and plan attrition are reported with their denominators, followed by executable-candidate J1-J5 attrition.

Integrity and replay

Requests, representations, ancestry edges, fitted programs, commitments, gate values and configuration hashes were retained. Replay reconstructed representations and recomputed J0-J5. Three fair-interface shard launches were interrupted after queue starvation produced transport timeouts but before any response or scientific result was returned; their logs were archived outside the formal namespace under a frozen operational amendment, and the unchanged shards were restarted sequentially. No completed shard was rerun and there were no outcome-quality exclusions.

AI assistance in research and writing

Phi-4 generated the original registered outputs described above. OpenAI Codex was used after the original confirmatory experiments to inspect artifacts, implement and orchestrate the separately frozen targeted sensitivity extension, recompute inference-free audits, verify literature metadata and assist drafting. It did not choose hypotheses using observed outcomes, alter historical source data or change the frozen evaluation rules. Human authors verified the cited metadata and remain responsible for originality, accuracy and integrity.

Data availability

Synthetic-world definitions, historical and extension result tables, raw call ledgers, manifests and replay artifacts are included in the project repository. The original confirmatory state and separate sensitivity namespace are identified by commits and tags. The submission release will archive an exact commit in a DOI-minting repository. No personal or restricted third-party data were used.

Code availability

Source code for generation, search, evaluation, offline attrition, statistics, figure generation, component audit and replay is included in the project repository. The release will identify the exact commit and archival DOI. Model weights are not redistributed; repository identifier, revision and runtime are reported above.

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Supplementary Information

Supplementary Methods

S1. Formal estimand

Let R0 denote a world's incumbent typed representation and H(R0) the frozen set of executable hypotheses admitted by its LanguageSpec. The incumbent oracle h0* minimizes observation loss over H(R0). A candidate representation R is an escape only when it is DSL-valid and $R \notin H(R0)$ under canonical membership checks. A world-level validated jump requires at least one of three candidates to pass J0-J5; semantic similarity and model judgment are excluded.

The jump success rate is the proportion of jump worlds with a validated candidate. The false-jump rate applies the identical decision to control worlds whose truth is in H(R0). Abductive precision is the number of validated candidates divided by candidates passing J0-J3. These quantities answer different questions and are never substituted for one another.

S2. Procedural families

AJ5 generated eight families: latent common cause, unification, hidden regimes, property-to-relation, state invention, coordinate transformation, causal ambiguity and meta-law. Each generator returned public observations, an incumbent representation and language, a hidden executable truth, a finite intervention domain and independent falsification cases. Generator validation checked deterministic output, truth redaction, exact incumbent observational fit, existence of a discriminating intervention and reachability of a valid escaped explanation.

Jump seeds were 10000-10049 per family and controls 20000-20024 per family. CJ5 used reconstruction seeds 30000-30049, controls 50000-50024, held-out jump seeds 40000-40099 and held-out controls 60000-60099. World generation used no LLM output.

S3. Typed representation and incumbent membership

A representation is a canonical graph of typed nodes and edges plus executable equations, dependencies, constraints, observability, arity, temporal indices and argument bindings. Canonical serialization sorts unordered fields before hashing. LanguageSpec.contains checks node kinds, edge/relation forms, equation families and other frozen admissibility constraints. J1 records at least one concrete membership failure and cannot be satisfied by renaming or paraphrase.

The incumbent oracle is exhaustive where the admissible grid is finite and bounded by a registered exact procedure otherwise. Any failure of J0 invalidates a world rather than removing it as difficult. No such confirmatory failure occurred.

S4. AJ5 conditions

- B0 direct LLM: directly emits a typed theory graph.
- B1 sample matched: high-temperature independent samples propose at most three typed operations.
- B2 fixed-space agent: representation remains R0; calls may reason and revise inside it.

- B3 attribute mutation: an external mechanism changes only values/equation attributes allowed by H(R0).
- B4 representation mutation: three proposals sampled with replacement from a nine-member typed portfolio.
- B5 full system: three structurally distinct portfolio proposals, archive accounting and deterministic falsification.

The portfolio contained latent, invariant, sign-contrast regime, additive relation, additive state, square-function, affine-context, causal-edge and transition variants. This high-level alignment motivated CJ5 and is not hidden as a limitation.

The P0/P1/P2 factorial reused the same reasoning code and world populations. Only the proposal source differed: LLM, external portfolio or oracle-correct representation. P2 did not receive fitted parameters, truth-program output or intervention outcomes.

S5. CJ5 generic primitive language

GENERIC_PRIMITIVE_SET_V1 contains 29 local operations: add/remove node, edge, function, equation, dependency, constraint and temporal index; reverse/retype/relabel; change observability or arity; bind/unbind; compose/decompose; merge/split; reify edge or node; and bounded copy/crossover. ADD_NODE creates an untyped Primitive; typing, observability, arity, temporal indexing, dependencies and each argument binding require separate records.

Every record binds the parent hash, child hash, operator, canonical arguments, seed and depth. The implementation rejects family names, truth, outcome fields, target distance and family-level aliases such as LATENTIZE, ADD_STATE, PROPERTY_TO_RELATION, ADD_REGIME, COMMON_CAUSE, META_LAW and COORDINATE_TRANSFORM.

S6. CJ5 search and controls

C3 traversed 48 branches of four local edits. Fixed allocations covered generic topologies for typed nodes, function composition, argument-bound functions, relations and reified edges. Ranking used only DSL validity, incumbent non-membership, observational compatibility, existence of an outcome-blind discriminating query, structural novelty and ancestry diversity.

C0 evaluated 192 within-space candidates. C2 evaluated 192 depth-one alternatives. C_rand drew 48 four-step paths and selected final candidates by seed-fixed structural hash. C_self supplied the identical manifest to Phi-4 and requested 16 plans in each of three slots; invalid plans consumed capacity and were not externally completed. C1 retained the AJ5 atomic portfolio as a reference. C5 supplied the target representation and deterministic truth compiler as a conditional ceiling. Because the operation semantics of C1 and C5 differ, they are shown separately in cost comparisons.

S7. Held-out family and lock

triadic_relation_reification uses three correlated observed inputs. The incumbent language fits a cubic rule on the observational support. The target represents an arity-three relation whose product mechanism separates under intervention. A valid construction requires edge reification, an arity change and separate argument bindings. No primitive produces the complete target.

The generator, structure and confirmatory seeds were excluded from AJ5 and CJ5 pilot inference. Only deterministic unit tests for redaction, oracle adequacy and constructive reachability preceded the lock. Known-family and control runs were terminal before the held-out unlock commit. Because AJ5 contained a binary property-to-relation family and the generic set already included reification, this is structural-family hold-out, not wholly concept-free invention.

S8. Candidate fitting and intervention selection

The shared fitter deterministically maps a valid representation and public observations to an executable program. It cannot read hidden truth, validation outcomes or family labels. For J3, the designer evaluates candidate and incumbent predictions over a finite public action set and selects their maximum absolute separation. It records action, predictions, candidate hash and split hash before querying the simulator. Model-proposed actions are retained only as diagnostics.

Thresholds were frozen: J0 and J2 observation MSE<=10-12; J3 separation>=0.5; J4 candidate intervention MSE < incumbent MSE-0.1; J5 candidate falsification MSE<=10-12 and < incumbent MSE-0.1. All six gates are conjunctive.

S9. Opportunity and compute accounting

Every condition-world cell had three final slots, two LLM calls per slot, one final candidate evaluation and one intervention commitment per slot, with 700 completion tokens per call. AJ5 therefore executed 32,400 prospectively specified calls plus 3,600 A6 calls. CJ5 executed 33,600 calls. Unused completion capacity was not reassigned. Actual calls, tokens, attempted/valid operations, candidate evaluations, interventions and latency are reported rather than claiming equality of realized token counts.

S10. Statistical analysis

The replicate was the world. AJ5 used 10,000 deterministic, family-stratified paired bootstrap replicates with seed 20260902. Seeds were resampled inside each family and conditions remained paired; the aggregate gave each family equal weight. Percentile 95% intervals were reported for rates and paired differences. One-sided empirical P values included the finite-replicate correction and eight primary comparisons were Holm adjusted.

CJ5 used the same stratified bootstrap for effect intervals and paired random sign-flip tests for registered beneficial contrasts. C3-C0 and C3-C2 formed the known-family primary multiplicity family; C3-C1, C3-C_rand and C3-C_self formed a secondary family. Held-out contrasts formed their own prospectively specified family. Two-sided Wilson intervals summarize JSR/FJR. No candidate-level pseudoreplication was used.

Post-hoc component audit

An inference-free audit replaced both C3 model calls with a valid empty explanation while preserving only the archived deterministic representation, fitted expression and maximum-separation intervention. Every world was regenerated and every commitment and J0-J5 verdict recomputed. All 2,400 candidate verdicts matched, retaining 500/500 jump successes and 0/300 control false jumps. The audit is explicitly post-hoc and changes attribution, not the registered comparison.

In the 400 known-family worlds, all 1,200 historical C_self responses were non-empty and the registered legacy parser extracted a JSON object, but the outer plans value was never a list. Thus 0/19,200 registered plan opportunities reached execution or structural evaluation. Every response reached the 700-token cap and strict whole-response JSON validity was 0/1,200. Incumbent fallback candidates inserted after empty self-search are excluded from proposal attrition. The analysis made no model calls.

Retained jump gain was rho_J=(JSR_C3-JSR_C0)/(JSR_C1-JSR_C0). It is undefined for a non-positive denominator. It compares observed rates and does not normalize unequal operator semantics.

S11. Exclusions, failures and replay

Parse errors, invalid graphs, non-exact fits, absent discriminating interventions and failed falsification were outcomes, not exclusions. A whole shard could be rerun only for a configuration/hash mismatch, unavailable server, missing/duplicate rows or infrastructure interruption, using identical seeds and model configuration. Confirmatory exclusions and shard reruns were both zero.

Replay rebuilt each selected representation from R0 and its ancestry, recomputed canonical hashes and fitted expressions, reconstructed the frozen intervention, and recomputed J0-J5. AJ5 reproduced 10,800/10,800 selected candidates; CJ5 reproduced 16,800/16,800 with 35,533 ancestry records. There were no mismatches.

Minimal targeted sensitivity protocol

Historical Phi-4 results were preserved on a dedicated tag and archive branch. The new protocol used a separate namespace and a commit-frozen panel selected by outcome-blind salted SHA-256 ranking. The same 12 existing seeds were applied to each of eight known families, yielding 96 paired worlds; no new generator or seed was introduced. A balanced supplied-representation positive control used five of those seeds per family (n=40).

The only new model conditions were Phi-4 8-bit C_self, DeepSeek matched C_self (reasoning_effort=none, 700 output tokens), DeepSeek native C_self (reasoning_effort=max, 4,096 output tokens) and DeepSeek supplied-representation positive control (reasoning_effort=max, 4,096 output tokens). Offline attrition crossed the prospectively frozen 25% pre-execution trigger, enabling one Phi-4 8-bit repair condition. The repair replaced, rather than augmented, a structurally invalid response and received only validator error classes; it could not receive semantic or simulator feedback.

The separately frozen PHI-BUDGET-SENSITIVITY reused the exact historical Phi-4 revision and 4-bit runtime. Prompt semantics, primitive vocabulary, known and held-out worlds, three candidate slots, 16 four-step plans per slot, representation opportunities, interventions, J0-J5 and all hidden-information boundaries were unchanged. Only generation.max_tokens increased from 700 to 2,048. The condition therefore tests sensitivity to the original completion cap; it is not compute-matched. Its source protocol was frozen at commit 4606413 and tag nmi-extension-v1-protocol-freeze. The completed known-family (n=400) and held-out (n=100) shards were incorporated before outcome analysis through amendment NMI-MIN-SENS-V1-AMENDMENT-002; the preselected 96-world panel is the primary paired comparison and the full shards are descriptive.

Each shard had to match its frozen seed panel, model identity, revision, call cardinality and artifact schema before receiving complete_verified status. Replay was locked until all five shards were verified. Analysis was then locked until all outputs replayed with zero mismatches. Reported inference consists of counts, Wilson intervals, paired world transitions, paired JSR differences and per-family descriptions. DeepSeek native is explicitly not compute-matched, and the Phi-4 sensitivity changes serving engine together with precision.

S12. Minimal sensitivity results and attrition

All legacy-interface autonomous fixed-panel conditions produced 0/96 successful worlds (Wilson 95% interval 0-3.8%): the historical Phi-4 4-bit slice, Phi-4 4-bit with the 2,048-token cap, Phi-4 8-bit, DeepSeek matched, DeepSeek native and Phi-4 8-bit with one validator-only repair. The five registered paired contrasts therefore contained 96 joint failures, no successes unique to either member and a paired JSR difference of 0.000. Each of the eight families contributed 12 worlds and had 0/12 in every legacy-interface autonomous condition; these family counts are descriptive.

The fixed-panel 2,048-token Phi-4 run produced 993/4,608 schema-valid plan opportunities but no executable plan. Across the complete descriptive known-family population, 4,642/19,200 opportunities were schema-valid and 9 executable. Those 9 plans occurred within a single selected slot, yielding one candidate, which failed J1. The held-out population produced 1,289/4,800 schema-valid and 0 executable opportunities. JSR was 0/400 known-family and 0/100 held-out, matching the historical world-level counts.

Phi-4 8-bit produced 4,608/4,608 JSON-extractable effective opportunities but 0 schema-valid because the extracted outer plans field was not a list. Its one-repair condition repeated the same error in every effective replacement opportunity. DeepSeek matched had 4,480 non-list plans errors and 128 missing plans among 4,608 opportunities. DeepSeek native exposed reasoning text in all 576 calls, but no separate answer reached the parser; 518 calls ended at the 4,096-token cap and 58 reported a stop finish reason.

The balanced DeepSeek supplied-representation control succeeded in 3/40 worlds (7.5%; Wilson 95% interval 2.6-19.9%): 1/5 coordinate-transformation and 2/5 hidden-regime worlds, with 0/5 in each other family. Of 120 model-authored expression-and-intervention candidates, 4 were parse-valid, executable and passed J1-J2; 3 also passed J3-J5. The other calls supplied reasoning text without a usable final answer; 116/120 reached the cap. This is a minimum positive control for occasional supplied-representation use, not a reliable ceiling.

The extension used 576 calls in each unrepaired n=96 C_self condition, 864 in the triggered repair and 120 in P2. Total completion tokens were 231,773 for Phi-4 8-bit, 237,792 for DeepSeek matched, 2,249,518 for DeepSeek native, 433,373 for repaired Phi-4, 485,312 for P2 and 528,976 for the fixed-panel Phi-budget condition. The service exposed reasoning text but not a separate reasoning-token count for DeepSeek. All five new shards and both Phi-budget populations replayed without mismatch: 2,772 candidate rows, zero mismatches and zero replay-time model calls.

S13. Fair-interface sensitivity protocol

An offline audit of the matched and native DeepSeek proposal calls made no model calls. It found that the registered self-composition prompt listed the available operators but not the exact argument names expected by the executor. All 288 matched proposal responses used at least one of the incompatible keys source, target or type; none was strict whole-response JSON. The legacy parser could extract an inner JSON object from an incomplete outer response, so object extraction was not evidence of a valid top-level plans object. Native proposal calls contained reasoning but no answer content. These findings motivated a single fair-interface sensitivity rather than a broader model factorial.

The fair condition reused the fixed 96-world panel, three slots, 16 four-step plans per slot, 28 available non-crossover primitives, deterministic fitter and intervention selector, and unchanged J0-J5. Each slot used two calls to the same served DeepSeek checkpoint. A native-reasoning call had a 4,096-token budget. A second reasoning_effort=none call received only the first call's deliberation, public world and exact parser-level syntax and had a separate 4,096-token answer budget. Its strict JSON schema fixed the outer object, 16-plan count, four-step depth, operator vocabulary and operator-specific argument keys and string types. It did not guarantee that referenced nodes or edges existed when used, that operations composed successfully, or that a resulting representation fit observations or separated intervention predictions.

No truth, target distance, fitted expression, intervention outcome, J3-J5 value or semantic validator feedback entered either call. The two-stage allocation preserved six calls per world by replacing the original proposal-plus-explanation calls; no additional representation opportunity was added. Protocol, configuration and code hashes were frozen at commit b6e1561 and annotated tag nmi-fair-interface-v1-protocol-freeze. An operational amendment at commit a65974f and tag nmi-fair-interface-v1-shard-freeze partitioned the eight families into four disjoint two-family shards while retaining concurrency four inside each runner. The server admitted at most four simultaneous sequences. Concurrent launch starved three shards behind that limit: each accumulated four 600-s timeout records but no returned response, candidate table, world table or summary. Those timeout-only directories were archived outside the formal namespace and excluded; a second operational amendment at commit 6bab5fb and tag nmi-fair-interface-v1-sequential-shards froze sequential execution of the unchanged shards. This changed wall time and scheduling only. A separate 31-call pre-amendment throughput pilot likewise produced no result tables and was excluded from all analyses.

S14. Fair-interface sensitivity results

The fair interface produced strict whole-response JSON, schema-valid operation names and parser-level argument types in 4,608/4,608 plan opportunities. Dynamic execution succeeded for 3,939/4,608 plans (85.5%), and 280/288 selected candidate slots contained at least one executable proposal. Cumulative selected-candidate attrition was 236/280 at J1, 236/280 at J2 and 21/280 at each of J3, J4 and J5. The 21 validated candidates yielded 15/96 successful worlds (15.6%; Wilson 95% interval 9.7-24.2%). Compared separately with the historical Phi-4 slice, matched DeepSeek and native DeepSeek, each paired table contained 81 joint failures, 15 successes unique to the fair interface, no successes unique to the reference and a paired JSR difference of +0.156.

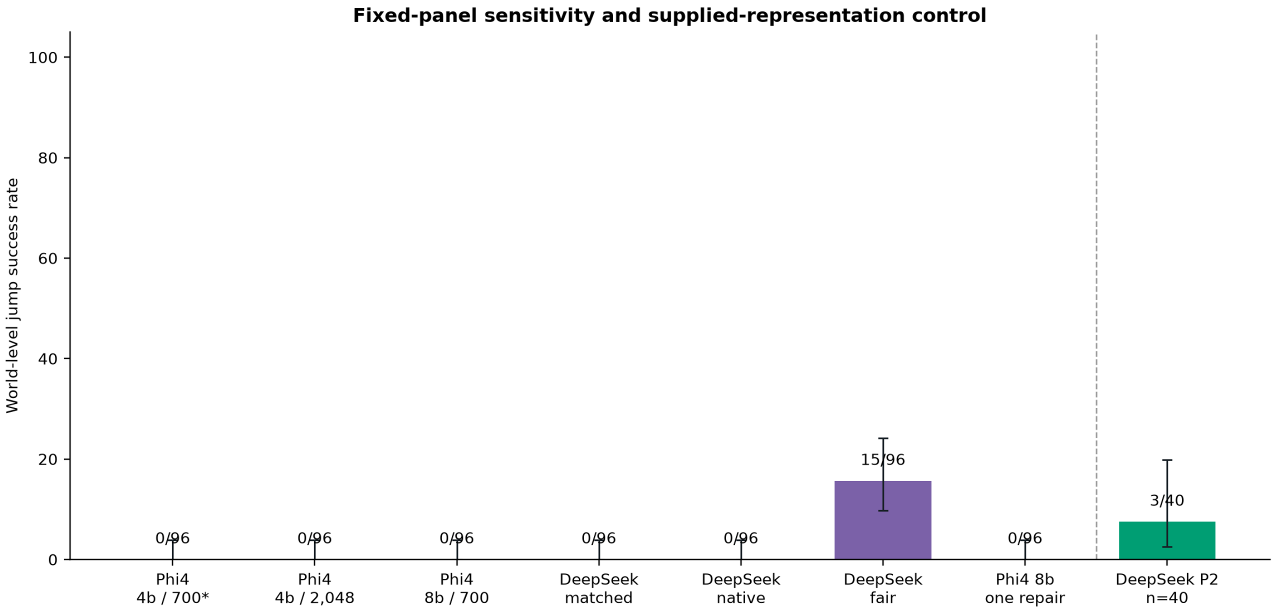
Success was family-concentrated: meta-law succeeded in 9/12 worlds and unification in 6/12; causal ambiguity, coordinate transformation, hidden regimes, latent common cause, property-to-relation and state invention each succeeded in 0/12. Family counts are descriptive. All 288 deliberation calls returned reasoning text and reached the 4,096-token cap. All 288 serialization calls returned strict schema-valid answers and stopped before their separate cap. Deliberation and serialization used 1,179,648 and 574,538 completion tokens, respectively; the service exposed no separate reasoning-token count. The formal condition used 576 calls and no transport retry.

Deterministic replay verified all 288 candidate rows with zero scientific, prompt or request mismatches and zero replay-time model calls. The initially frozen verifier nevertheless reported 576 metadata mismatches because it expected an engine field on each call-ledger row, whereas the frozen runner stores engine metadata in each shard summary's stage manifests. A post-hoc verifier correction preserved the initial report and frozen verifier hash, verified model identity, revision, quantization, engine, engine version and generation settings across all four shard manifests, and omitted only the nonexistent per-call engine comparison. No inference artifact or scientific result changed.

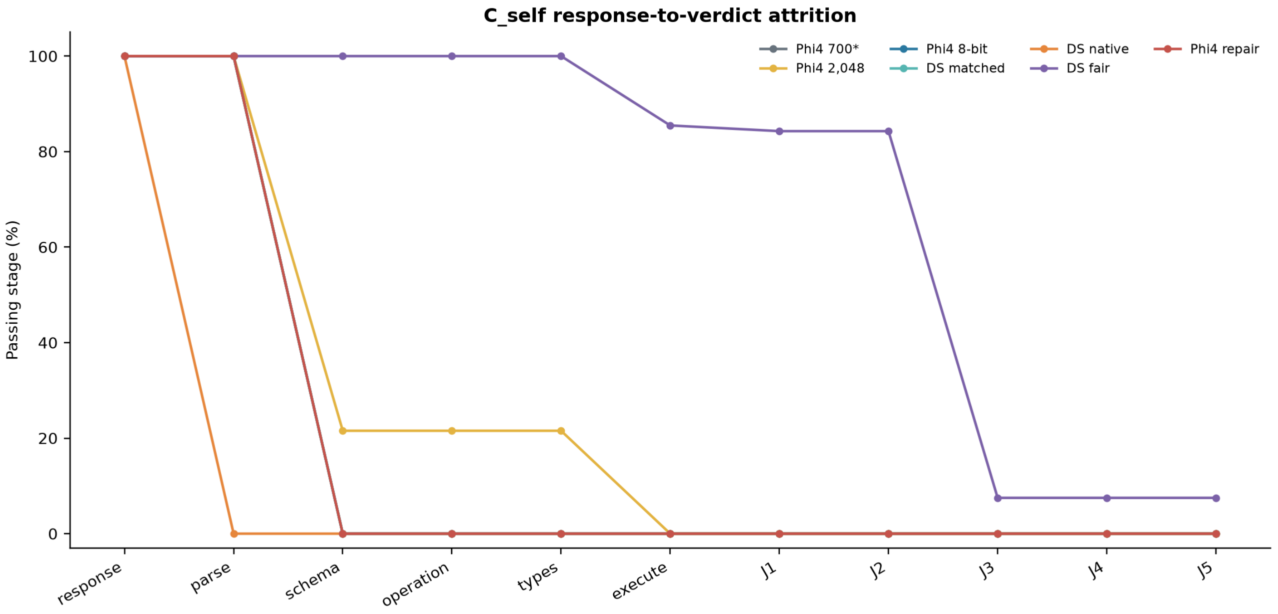
S15. Software and environment

The repository records package versions, configuration hashes, prompt identifiers, model revision, vLLM image digest, GPU class and random seeds in reproducibility manifests. Tests cover world determinism, public-field redaction, DSL membership, primitive legality, budget invariants, reachability, statistics and replay. The publication phase did not call the model or generate new experimental rows.

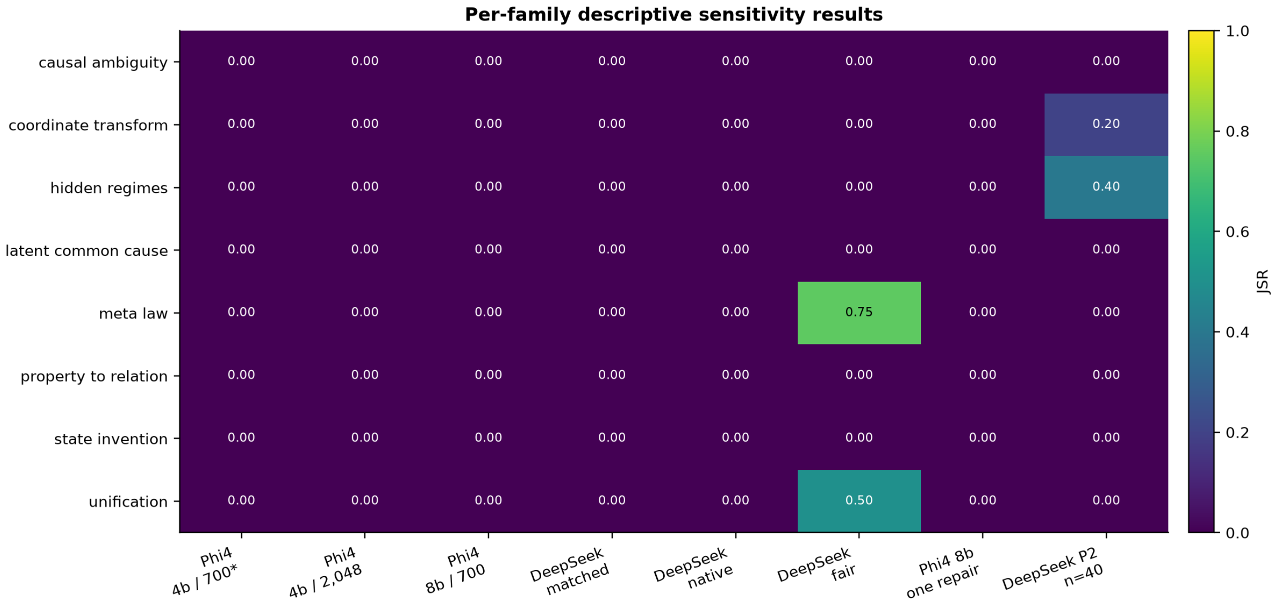
Targeted sensitivity source data



Extended Data Figure 8a | Exact world-level sensitivity results and Wilson 95% intervals, including the separately frozen fair-interface condition. The asterisk identifies the fixed historical slice of the original n=400 confirmation; the supplied-representation control uses a distinct balanced n=40 subset.



Extended Data Figure 8b | Response-to-verdict attrition, including the grammar-valid fair-interface cascade. Denominators and units change at the executable boundary and are reported in the accompanying source-data tables.



Extended Data Figure 8c | Per-family descriptive sensitivity results. Fair-interface successes were confined to meta-law and unification; no family-level or candidate-level significance test was performed.

Sensitivity result table

Condition	Population	Success	JSR	Wilson 95% CI
historical phi4 4bit cself	original confirmatory n=400, fixed paired subset shown	0/96	0.0%	0.0-3.8%
phi4 4bit budget cself	previously frozen n=400 budget sensitivity, fixed paired subset shown	0/96	0.0%	0.0-3.8%
phi8 cself	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
deepseek matched cself	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
deepseek native cself	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
phi8 cself repair	new fixed n=96 sensitivity panel	0/96	0.0%	0.0-3.8%
deepseek p2	new balanced n=40 positive-control subset	3/40	7.5%	2.6-19.9%
deepseek fair interface cself	fixed n=96 sensitivity panel	15/96	15.6%	9.7-24.2%

Paired world-level transitions

Reference	Comparison	Both fail	Both pass	New only	Old only	JSR difference
historical phi4 4bit cself	phi4 4bit budget cself	96	0	0	0	+0.000
historical phi4 4bit cself	phi8 cself	96	0	0	0	+0.000
historical phi4 4bit cself	deepseek matched cself	96	0	0	0	+0.000
deepseek matched cself	deepseek native cself	96	0	0	0	+0.000
phi8 cself	phi8 cself repair	96	0	0	0	+0.000
historical phi4 4bit cself	deepseek fair interface cself	81	0	15	0	+0.156
deepseek matched cself	deepseek fair interface cself	81	0	15	0	+0.156
deepseek native cself	deepseek fair interface cself	81	0	15	0	+0.156

Complete cumulative gate attrition

Condition	Unit	Stage	Passed	Denominator	Rate
historical phi4 4bit cself	proposal_response	response_returned	1200	1200	100.0%
historical phi4 4bit cself	proposal_response	parse_valid	1200	1200	100.0%
historical phi4 4bit cself	proposal_response	schema_valid	0	1200	0.0%
historical phi4 4bit cself	proposal_response	operation_valid	0	1200	0.0%
historical phi4 4bit cself	proposal_response	argument_type_valid	0	1200	0.0%
historical phi4 4bit cself	proposal_response	executable	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J1	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J2	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J3	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J4	0	1200	0.0%
historical phi4 4bit cself	proposal_response	J5	0	1200	0.0%
phi8 cself	effective_plan_opportunity	request_returned	4608	4608	100.0%
phi8 cself	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
phi8 cself	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
phi8 cself	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%
phi8 cself	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
phi8 cself	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
phi8 cself	effective_plan_opportunity	executable	0	4608	0.0%
phi8 cself	effective_plan_opportunity	representation_constructed	0	4608	0.0%
phi8 cself	executable_self_proposed_candidate	J1	0	0	0.0%
phi8 cself	executable_self_proposed_candidate	J2	0	0	0.0%
phi8 cself	executable_self_proposed_candidate	J3	0	0	0.0%
phi8 cself	executable_self_proposed_candidate	J4	0	0	0.0%
phi8 cself	executable_self_proposed_candidate	J5	0	0	0.0%
deepseek matched cself	effective_plan_opportunity	request_returned	4608	4608	100.0%
deepseek matched cself	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
deepseek matched cself	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
deepseek matched cself	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%

Condition	Unit	Stage	Passed	Denominator	Rate
deepseek matched cself	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
deepseek matched cself	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
deepseek matched cself	effective_plan_opportunity	executable	0	4608	0.0%
deepseek matched cself	effective_plan_opportunity	representation_constructed	0	4608	0.0%
deepseek matched cself	executable_self_proposed_candidate	J1	0	0	0.0%
deepseek matched cself	executable_self_proposed_candidate	J2	0	0	0.0%
deepseek matched cself	executable_self_proposed_candidate	J3	0	0	0.0%
deepseek matched cself	executable_self_proposed_candidate	J4	0	0	0.0%
deepseek matched cself	executable_self_proposed_candidate	J5	0	0	0.0%
deepseek native cself	effective_plan_opportunity	request_returned	4608	4608	100.0%
deepseek native cself	effective_plan_opportunity	non_empty_answer	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	json_parse_valid	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	executable	0	4608	0.0%
deepseek native cself	effective_plan_opportunity	representation_constructed	0	4608	0.0%
deepseek native cself	executable_self_proposed_candidate	J1	0	0	0.0%
deepseek native cself	executable_self_proposed_candidate	J2	0	0	0.0%
deepseek native cself	executable_self_proposed_candidate	J3	0	0	0.0%
deepseek native cself	executable_self_proposed_candidate	J4	0	0	0.0%
deepseek native cself	executable_self_proposed_candidate	J5	0	0	0.0%
phi8 cself repair	effective_plan_opportunity	request_returned	4608	4608	100.0%
phi8 cself repair	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
phi8 cself repair	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
phi8 cself repair	effective_plan_opportunity	plan_schema_valid	0	4608	0.0%

Condition	Unit	Stage	Passed	Denominator	Rate
phi8 cself repair	effective_plan_opportunity	operation_names_valid	0	4608	0.0%
phi8 cself repair	effective_plan_opportunity	argument_types_valid	0	4608	0.0%
phi8 cself repair	effective_plan_opportunity	executable	0	4608	0.0%
phi8 cself repair	effective_plan_opportunity	representation_constructed	0	4608	0.0%
phi8 cself repair	executable_self_proposed_candidate	J1	0	0	0.0%
phi8 cself repair	executable_self_proposed_candidate	J2	0	0	0.0%
phi8 cself repair	executable_self_proposed_candidate	J3	0	0	0.0%
phi8 cself repair	executable_self_proposed_candidate	J4	0	0	0.0%
phi8 cself repair	executable_self_proposed_candidate	J5	0	0	0.0%
phi4 4bit budget cself	effective_plan_opportunity	request_returned	4608	4608	100.0%
phi4 4bit budget cself	effective_plan_opportunity	non_empty_answer	4608	4608	100.0%
phi4 4bit budget cself	effective_plan_opportunity	json_parse_valid	4608	4608	100.0%
phi4 4bit budget cself	effective_plan_opportunity	plan_schema_valid	993	4608	21.5%
phi4 4bit budget cself	effective_plan_opportunity	operation_names_valid	993	4608	21.5%
phi4 4bit budget cself	effective_plan_opportunity	argument_types_valid	993	4608	21.5%
phi4 4bit budget cself	effective_plan_opportunity	executable	0	4608	0.0%
phi4 4bit budget cself	effective_plan_opportunity	representation_constructed	0	4608	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J1	0	0	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J2	0	0	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J3	0	0	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J4	0	0	0.0%
phi4 4bit budget cself	executable_self_proposed_candidate	J5	0	0	0.0%
deepseek p2	model_authored_expression_and_intervention_candidate	parse_valid	4	120	3.3%
deepseek p2	model_authored_expression_and_intervention_candidate	executable	4	120	3.3%
deepseek p2	model_authored_expression_and_intervention_candidate	J1	4	120	3.3%
deepseek p2	model_authored_expression_and_intervention_candidate	J2	4	120	3.3%

Condition	Unit	Stage	Passed	Denominator	Rate
deepseek p2	model_authored_expression_and_intervention_candidate	J3	3	120	2.5%
deepseek p2	model_authored_expression_and_intervention_candidate	J4	3	120	2.5%
deepseek p2	model_authored_expression_and_intervention_candidate	J5	3	120	2.5%

Model-call and compute ledger

Condition	Calls	Attempts	Prompt tokens	Completion tokens	Reasoning text	Latency (s)
phi8 cself	576	576	704376	231773	0	13971.8
deepseek matched cself	576	576	786804	237792	0	10636.3
deepseek native cself	576	576	839796	2249518	576	156547.1
phi8 cself repair	864	864	1254663	433373	0	26462.2
deepseek p2	120	120	183891	485312	120	38836.1
phi4 4bit budget cself	576	576	704376	528976	0	40446.1

Reasoning-token counts are reported only when exposed by the serving API; reasoning-text availability is not converted into an inferred token count.

Full Phi-4 completion-budget sensitivity

Condition	Population	Success	JSR	Wilson 95% CI
historical phi4 4bit cself full	original confirmatory n=400	0/400	0.0%	0.0-1.0%
phi4 4bit budget cself full	budget sensitivity n=400	0/400	0.0%	0.0-1.0%
historical phi4 4bit cself heldout	original held-out n=100	0/100	0.0%	0.0-3.7%
phi4 4bit budget cself heldout	budget sensitivity held-out n=100	0/100	0.0%	0.0-3.7%

Full paired completion-budget transitions

Reference	Comparison	Both fail	Both pass	New only	Old only	Difference
historical phi4 4bit cself full	phi4 4bit budget cself full	400	0	0	0	+0.000
historical phi4 4bit cself heldout	phi4 4bit budget cself heldout	100	0	0	0	+0.000

Full Phi-4 completion-budget gate attrition

Population	Unit	Stage	Passed	Denominator	Rate
phi4 4bit budget cself full	effective_plan_opportunity	request_returned	19200	19200	100.0%
phi4 4bit budget cself full	effective_plan_opportunity	non_empty_answer	19200	19200	100.0%
phi4 4bit budget cself full	effective_plan_opportunity	json_parse_valid	19200	19200	100.0%
phi4 4bit budget cself full	effective_plan_opportunity	plan_schema_valid	4642	19200	24.2%
phi4 4bit budget cself full	effective_plan_opportunity	operation_names_valid	4642	19200	24.2%
phi4 4bit budget cself full	effective_plan_opportunity	argument_types_valid	4642	19200	24.2%
phi4 4bit budget cself full	effective_plan_opportunity	executable	9	19200	0.0%
phi4 4bit budget cself full	effective_plan_opportunity	representation_constructed	9	19200	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J1	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J2	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J3	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J4	0	1	0.0%
phi4 4bit budget cself full	executable_self_proposed_candidate	J5	0	1	0.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	request_returned	4800	4800	100.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	non_empty_answer	4800	4800	100.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	json_parse_valid	4800	4800	100.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	plan_schema_valid	1289	4800	26.9%
phi4 4bit budget cself heldout	effective_plan_opportunity	operation_names_valid	1289	4800	26.9%
phi4 4bit budget cself heldout	effective_plan_opportunity	argument_types_valid	1289	4800	26.9%
phi4 4bit budget cself heldout	effective_plan_opportunity	executable	0	4800	0.0%
phi4 4bit budget cself heldout	effective_plan_opportunity	representation_constructed	0	4800	0.0%
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J1	0	0	0.0%
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J2	0	0	0.0%
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J3	0	0	0.0%
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J4	0	0	0.0%

Population	Unit	Stage	Passed	Denominator	Rate
phi4 4bit budget cself heldout	executable_self_proposed_candidate	J5	0	0	0.0%

Full Phi-4 completion-budget compute ledger

Population	Calls	Attempts	Prompt tokens	Completion tokens	Latency (s)
phi4 4bit budget cself full	2400	2400	2921915	2216914	169361.1
phi4 4bit budget cself heldout	600	600	696489	546374	41524.5

Fair-interface DeepSeek sensitivity

Family	Success	JSR	Wilson 95% CI
causal ambiguity	0/12	0.0%	0.0-24.2%
coordinate transform	0/12	0.0%	0.0-24.2%
hidden regimes	0/12	0.0%	0.0-24.2%
latent common cause	0/12	0.0%	0.0-24.2%
meta law	9/12	75.0%	46.8-91.1%
property to relation	0/12	0.0%	0.0-24.2%
state invention	0/12	0.0%	0.0-24.2%
unification	6/12	50.0%	25.4-74.6%

Fair-interface compute ledger

Stage	Calls	Prompt tokens	Completion tokens	Cap hits	Latency (s)
deliberation	288	721860	1179648	288	87421.1
serialization	288	1914226	574538	0	23363.1

Fair-interface cumulative attrition

Unit	Stage	Passed	Denominator	Rate
plan opportunity	deliberation_returned	4608	4608	100.0%
plan opportunity	serialization_returned	4608	4608	100.0%
plan opportunity	strict_whole_response_json	4608	4608	100.0%
plan opportunity	json_parse_valid	4608	4608	100.0%
plan opportunity	plan_schema_valid	4608	4608	100.0%
plan opportunity	operation_names_valid	4608	4608	100.0%
plan opportunity	argument_types_valid	4608	4608	100.0%
plan opportunity	executable	3939	4608	85.5%
plan opportunity	representation_constructed	3939	4608	85.5%
executable selected candidate	J1	236	280	84.3%
executable selected candidate	J2	236	280	84.3%
executable selected candidate	J3	21	280	7.5%
executable selected candidate	J4	21	280	7.5%
executable selected candidate	J5	21	280	7.5%